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All That Glitters Is Not Arbitrage*

Tracking Error, Mean Reversion and the Vanishing Edge in Tokenised Gold

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Abstract

PAXG and XAUT are blockchain tokens each redeemable for one troy ounce of physical gold. In a frictionless world both would trade at the gold price, and therefore at the same price as each other. Using daily data from February 2020 to October 2026, I measure how far reality departs from that ideal and whether anyone could profit from the difference. Against COMEX gold futures, both tokens show a daily-return beta of roughly 0.80 and annualised tracking error of 11–13%, figures I argue are inflated by asynchronous closing times and futures carry rather than by genuine mispricing. The spread between the two tokens is more revealing. It is strongly stationary (ADF $p \approx 4 \times 10^{-6}$) and, modelled as an Ornstein–Uhlenbeck process, has a half-life of just 2.45 days. A simple z-score relative-value rule earns a full-sample Sharpe ratio of 1.30 after 10 bps per-leg costs. That headline, however, is almost entirely historical. Splitting the sample shows the net Sharpe ratio falling from 1.88 (2020–22) to 0.93 (2023–24) to −0.02 (2025–26), as spread volatility collapses from 92 to 28 bps and the break-even trading cost falls from 37 to 10 bps per leg. The mean reversion survives; the profit does not. Tokenised gold has, in effect, been arbitraged into efficiency.

Keywords: tokenisation; real-world assets; gold-backed tokens; statistical arbitrage; Ornstein–Uhlenbeck process; market efficiency; alpha decay

JEL classification: G12, G14, C22

* This is an independent working paper. It has not been peer reviewed, and nothing in it is investment advice. The strategy it describes lost money after costs over the most recent 22 months, which should settle any temptation. No gold bars were moved, shorted or harmed in its production.

The author is a first-year undergraduate and a Research Analyst on the Academic Research Team of the LSESU FinTech Society, whose current project on tokenised real-world assets motivated this note. The views and all errors are the author's own. Replication code is available on the author's GitHub.

1 Introduction — *or, why two ounces of gold should cost the same*

Tokenisation promises to put real-world assets (RWAs) on a blockchain: treasury bills, property, private credit and, most simply of all, gold. The pitch is compelling. A token that represents a claim on a physical bar can be traded around the clock, split into tiny fractions and settled in minutes rather than days. Whether the pitch is delivered, though, depends on one unglamorous question: *does the token actually trade at the value of the thing it represents?* A tokenised asset that drifts away from its underlying is not a better version of the asset. It is a new asset with its own risks attached.

Gold is the cleanest possible laboratory for this question. The two largest gold tokens, Paxos Gold (PAXG) and Tether Gold (XAUT), are each backed by one troy ounce of London Good Delivery gold held in custody, so their fundamental value is unambiguous.[1] If markets were frictionless, each token would equal the spot gold price at every instant, and the difference between the two tokens would be identically zero. Any persistent deviation is therefore a direct, model-free measure of friction: imperfect arbitrage, segmented venues, redemption costs, or simply investor inattention.

This paper asks three questions. First, how accurately does each token track gold (Section 4.1)? Second, when PAXG and XAUT drift apart, how quickly does the gap close (Section 4.2)? Third, could a trader have profited from that closing after realistic costs, and has that opportunity persisted (Sections 4.3 and 5)? The answers are, respectively: *better than the raw numbers suggest*, *very quickly*, and *yes, but not any more*.

The third answer is the paper's main contribution. Relative-value strategies on near-identical securities are a well-studied corner of finance (Gatev, Goetzmann and Rouwenhorst, 2006; Avellaneda and Lee, 2010), and it is equally well established that anomalies tend to shrink once they are noticed (McLean and Pontiff, 2016) or as markets mature (Lo, 2004). Tokenised gold offers an unusually crisp, real-time example of that process. Over six and a half years, the mispricing between two economically identical tokens shrank until the profit from trading it was smaller than the cost of trading it.

2 Data — *in which weekends cause trouble*

All prices are daily closes in US dollars from Yahoo Finance, retrieved with the open-source `yfinance` library. The tokens are `PAXG-USD` and `XAUT-USD`, and the gold benchmark is the continuous front-month COMEX gold futures contract, `GC=F`. Table 1 summarises coverage. The common sample for the token pair begins on 7 February 2020, the first date with an XAUT price, and ends on 21 October 2026.

Table 1. Data coverage.

Series	*Source ticker*	*First date*	*Last date*	*Observations*	*Trading calendar*
Paxos Gold	`PAXG-USD`	1 Jan 2020	21 Oct 2026	2,486	7 days a week
Tether Gold	`XAUT-USD`	7 Feb 2020	21 Oct 2026	2,449	7 days a week
Gold futures	`GC=F`	2 Jan 2020	21 Oct 2026	1,713	Exchange weekdays

Two features of the data matter for everything that follows. First, the tokens trade continuously, while futures trade only on exchange days. Token–gold comparisons therefore use only the 1,713 (PAXG) and 1,688 (XAUT) dates on which both prices exist, so that a "daily" return spanning a weekend is measured over the same interval for both series. Second, the two kinds of closing price are not taken at the same moment. Yahoo's crypto closes are stamped at 00:00 UTC, whereas COMEX settlement occurs in the New York afternoon. As Section 4.1 shows, this asynchronicity does real

damage to any naïve tracking statistic. The PAXG–XAUT comparison, by contrast, is perfectly synchronous: both tokens trade on the same venues, every day, and are stamped at the same instant.

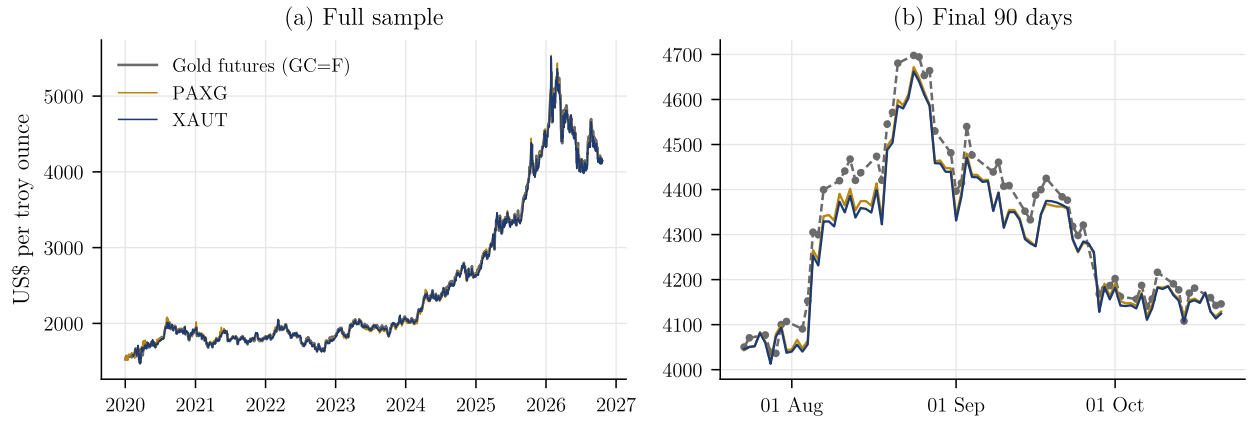


Figure 1. Gold futures and the two gold tokens. Panel (a): the full sample, over which gold futures rose from about \$1,520 to a peak of about \$5,320 in late January 2026. At this scale the three series are indistinguishable. Panel (b): the final 90 days. Dots mark futures settlements, and the dashed segments bridge weekends. Both tokens sit persistently, if slightly, below the futures price.

3 Methodology

3.1 Tracking accuracy

For token $i \in \{\text{PAXG}, \text{XAUT}\}$ with price P_t^i and gold futures price G_t, the premium is

$$\pi_t^i = \frac{P_t^i}{G_t} - 1. \tag{1}$$

With log returns $r_t^i = \ln P_t^i - \ln P_{t-1}^i$ measured between consecutive common dates, annualised tracking error is

$$\text{TE}^i = \sqrt{252}\ \text{sd}\left(r_t^i - r_t^G\right), \tag{2}$$

and co-movement is summarised by the ordinary least squares regression

$$r_t^i = \alpha^i + \beta^i\, r_t^G + \varepsilon_t^i, \tag{3}$$

for which perfect tracking implies $\beta^i = 1$ and $R^2 = 1$.

3.2 The spread and its mean reversion

The object of interest is the log spread between the two tokens,

$$s_t = \ln P_t^{\text{PAXG}} - \ln P_t^{\text{XAUT}}, \tag{4}$$

which is approximately the percentage price gap and should equal zero for economically identical claims. Stationarity is tested with the augmented Dickey–Fuller (ADF) test (Dickey and Fuller, 1979), using a constant and AIC lag selection. The speed of reversion comes from the regression

$$\Delta s_t = a + b\, s_{t-1} + \eta_t. \tag{5}$$

Equation (5) is the exact daily discretisation of an Ornstein–Uhlenbeck process (Uhlenbeck and Ornstein, 1930),

$$\mathrm{d}X_t = \theta\,(\mu - X_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \tag{6}$$

whose solution implies $E[X_{t+1} - \mu \mid X_t] = e^{-\theta}(X_t - \mu)$. Matching coefficients gives $1 + b = e^{-\theta}$. Hence

$$\theta = -\ln(1+b), \qquad t_{1/2} = \frac{\ln 2}{\theta}, \tag{7}$$

where the half-life $t_{1/2}$ is the expected number of days for a deviation from μ to halve.[2]

3.3 A relative-value trading rule

Let m_t and σ_t be the 30-day rolling mean and standard deviation of s_t, and define the z-score $z_t = (s_t - m_t)/\sigma_t$. The signal $g_t \in \{-1, 0, 1\}$ follows a simple state machine. From flat, it goes long the spread (buy PAXG, sell XAUT) when $z_t < -2$ and short the spread when $z_t > 2$. An open position is closed when z_t crosses back through zero. To rule out look-ahead bias, the position held on day t is yesterday's signal, $q_t = g_{t-1}$. Daily net returns per unit notional in each leg are

$$R_t = q_t \, \Delta s_t \; - \; 2c \; \mid q_t - q_{t-1} \mid , \tag{8}$$

with a cost of $c = 10$ bps per leg, so a complete round trip (two legs in, two legs out) costs 40 bps. Performance is summarised by the annualised Sharpe ratio $\mathrm{SR} = \sqrt{365}\ \bar{R}/\mathrm{sd}(R)$, using 365 days because the tokens never close, and by the maximum drawdown of the cumulative equity curve $E_t = \exp\Big(\sum_{u \le t} R_u \Big)$. Finally, the *break-even cost* is the per-leg cost at which the strategy's cumulative net return is exactly zero:

$$c^* = \frac{\sum_t q_t \, \Delta s_t}{\sum_t 2 \ \mid q_t - q_{t-1} \mid}. \tag{9}$$

The parameters (a 30-day window, ±2 entry and 0 exit) are textbook defaults, chosen before looking at the results and never tuned. This is a deliberate choice: an optimised rule would flatter the in-sample numbers and tell us less.

4 Results

4.1 Tracking against gold: worse than it looks, better than it is

Table 2 reports the tracking statistics. Taken at face value they are unflattering. Both tokens deviate from gold futures by more than 0.5% on roughly half of all days, carry annualised tracking error above 10%, and have a daily beta of only 0.80. Were these numbers an accurate portrait, gold tokens would be a rather poor way to own gold.

Two well-understood measurement effects explain much of this. The first is *non-synchronous trading*. Because token closes and futures settlements are recorded hours apart, part of each day's gold move appears in the token's return for the following day instead. Measurement error of this kind biases a regression slope towards zero, the classic problem identified by Scholes and Williams (1977) for thinly traded stocks. Figure 1(a) is the simplest rebuttal to a literal reading of $\widehat{\beta} = 0.8$: had the tokens really moved only 80% as much as gold, they could not have kept pace with a near-tripling of the gold price.

The second is *futures carry*. The benchmark is a futures contract, not spot gold. Under cost-of-carry pricing, $F_t \approx S_t \, e^{(r+u-y)\tau}$, so when interest rates r are high the futures price sits above spot. A token that tracks spot perfectly would then show a *negative* premium. Figure 2 is consistent with this. Premiums are close to zero or slightly positive before 2022, then turn persistently negative after the US Federal Reserve began raising rates in March 2022, reaching their lowest levels during 2023–24. Carry is a common factor, however, and cannot explain the differences between the tokens: over 2023–

24, PAXG traded on average 0.46 percentage points cheaper than XAUT. That token-specific component is precisely what the spread in Section 4.2 isolates.

Table 2. Tracking statistics against COMEX gold futures (common trading dates only).

	PAXG	*XAUT*
Common observations	1,713	1,688
Mean premium, $\bar{\pi}$	−0.26%	−0.35%
Standard deviation of premium	1.05%	0.75%
Share of days with $\lvert \pi_t \rvert > 0.5\%$	57.9%	46.7%
Annualised tracking error, eq. (2)	12.89%	10.64%
Beta, $\widehat{\beta}$, eq. (3)	0.797	0.802
R^2	0.607	0.708
Mean premium by sub-period		
2020–2022	+0.22%	−0.13%
2023–2024	−0.95%	−0.49%
2025–2026	−0.31%	−0.57%

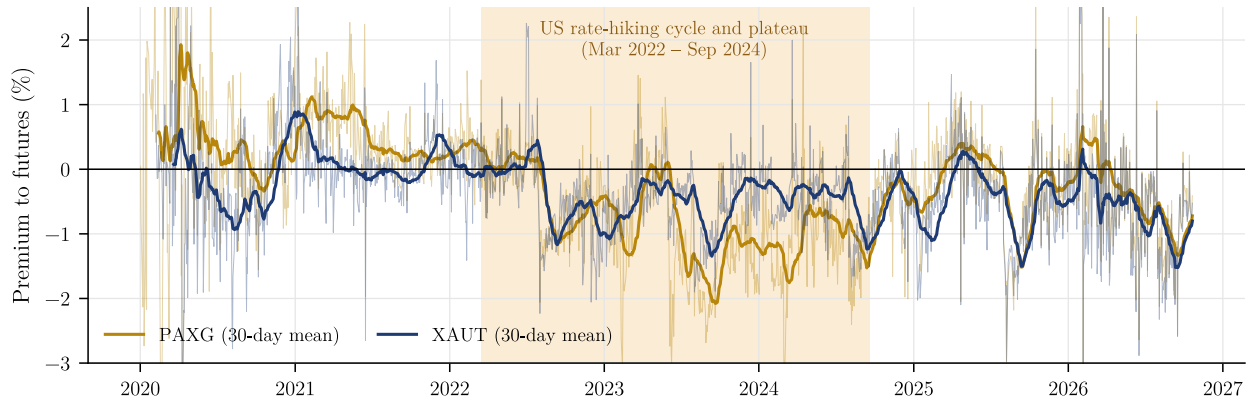


Figure 2. Premium of each token to gold futures, eq. (1). Thin lines are daily values and thick lines are 30-day means. The shaded window marks the US rate-hiking cycle and the 5.25–5.50% plateau (March 2022 to September 2024). The vertical axis is clipped at −3% and +2.5%, which truncates a handful of extreme observations in March 2020.

4.2 The PAXG–XAUT spread: a gap with a short memory

Comparing the tokens with each other removes both measurement problems at once: the two closes are synchronous, and there is no carry between two claims on the same physical asset. Figure 3(a) plots the spread. It is anything but zero, but it is clearly tethered. In the token markets' infancy in 2020 it routinely exceeded ±200 bps and briefly reached ±500 bps. By 2025–26 it rarely strays beyond ±50 bps. Its distribution, in Figure 3(b), is sharply peaked and heavy-tailed, with an excess kurtosis of 7.4. On most days the gap is small, and on a few days it is very large indeed, which is exactly the profile a relative-value trader hopes for.

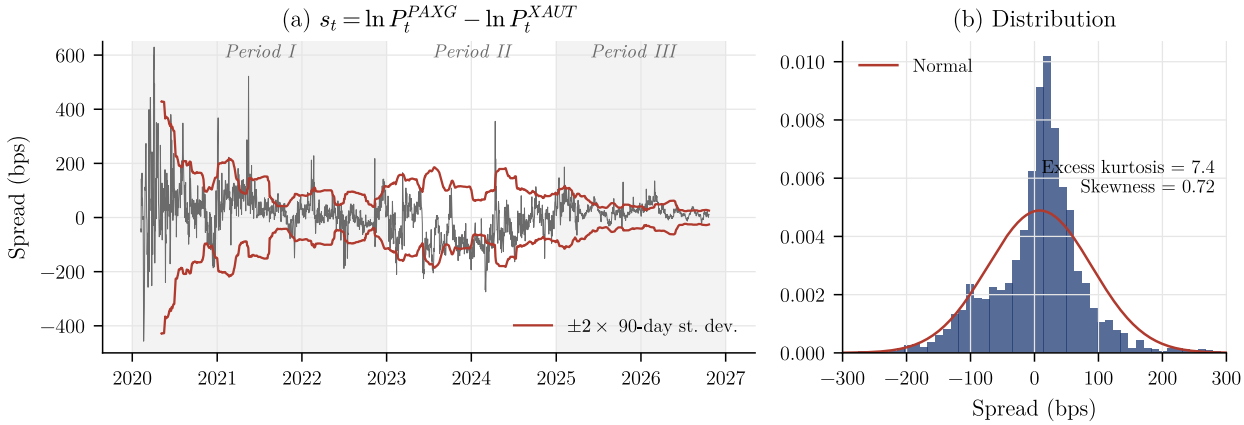


Figure 3. Panel (a): the PAXG–XAUT log spread in basis points with a ±2 rolling 90-day standard-deviation envelope. Shading marks the three sub-periods used in Section 5. Panel (b): the full-sample distribution against a normal density with the same mean and variance. Kurtosis and skewness are computed on the full sample.

Table 3. Mean-reversion estimates for the PAXG–XAUT spread, full sample (7 February 2020 to 21 October 2026, 2,449 days).

Statistic	*Estimate*	*Interpretation*
Mean spread	8.1 bps	PAXG marginally dearer on average
Standard deviation of spread	81.6 bps	
ADF statistic	−5.36	Unit root rejected ($p = 4.0 \times 10^{-6}$)
AR(1) slope, $\widehat{b}$, eq. (5)	−0.247 (0.013)	About 25% of a gap closes each day
OU speed, $\widehat{\theta}$, eq. (7)	0.283 per day	
Half-life, $t_{1/2}$	**2.45 days**	**A 100 bps gap is expected to be 50 bps within 2.5 days**

Standard error in parentheses (OLS).

Table 3 formalises the picture. The ADF test rejects a unit root at any conventional level. The slope in eq. (5) implies that roughly a quarter of any deviation is eliminated every day, an Ornstein–Uhlenbeck speed of $\widehat{\theta} = 0.283$ and a half-life of 2.45 days. Figure 4 shows both the regression and the decay it implies. By the standards of financial spreads this is fast. Avellaneda and Lee (2010), for instance, keep only US-equity residuals whose estimated mean-reversion time is below about 30 trading days. The gold-token gap closes in days, not weeks.

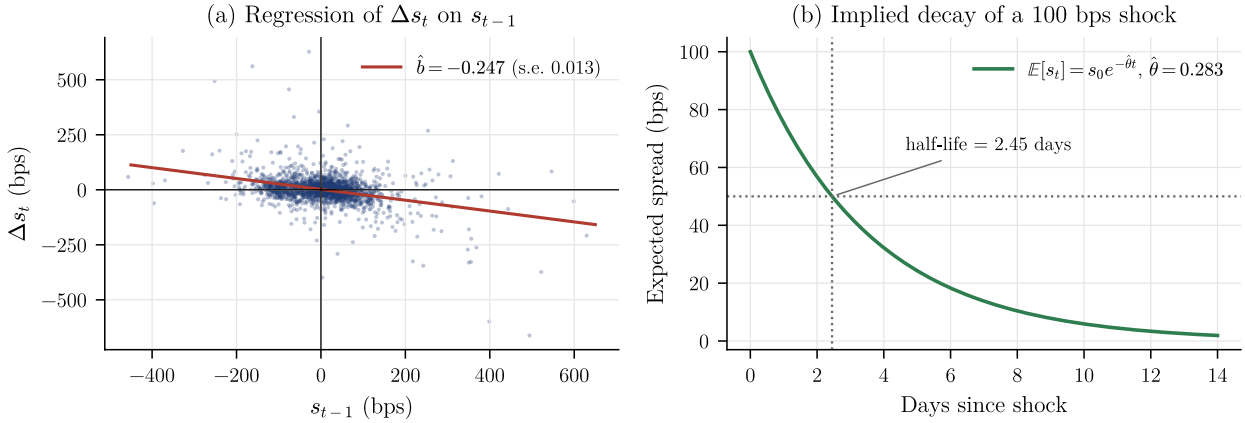


Figure 4. Panel (a): daily change in the spread regressed on its lagged level, eq. (5). The negative slope is the signature of mean reversion. Panel (b): the expected path of a 100 bps deviation implied by the estimated Ornstein–Uhlenbeck speed, with the half-life marked.

4.3 Trading the gap

Table 4 and Figure 5 report the trading rule of Section 3.3 over the full sample. Before costs, it earns 15.3% a year at 6.7% volatility, a Sharpe ratio of 2.13. After 10 bps per leg, it still earns 9.0% a year with a Sharpe ratio of 1.30 and a maximum drawdown of only −4.0%, across 95 round trips held for an average of 7.6 days. Costs absorb more than half of the gross profit, and the break-even cost of 25 bps per leg leaves a comfortable-looking margin over the assumed 10 bps.

Table 4. Performance of the z-score relative-value rule, full sample.

	Before costs	*After costs (10 bps per leg)*
Total return	160.3%	78.0%
Annualised return	15.32%	8.97%
Annualised volatility	6.70%	6.63%
Sharpe ratio	**2.13**	**1.30**
Maximum drawdown	−3.96%	−3.96%
Round trips	95	95
Share of days in a position	29.6%	29.6%
Average gross profit per round trip	100.7 bps	
Break-even cost per leg, c^*, eq. (9)	25.2 bps	

Returns are per unit of notional in each leg, with no leverage, funding or borrow costs. The before-cost and after-cost drawdowns coincide because no position was opened or closed between the peak and the trough of the worst episode (early 2021), so no costs were incurred within it.

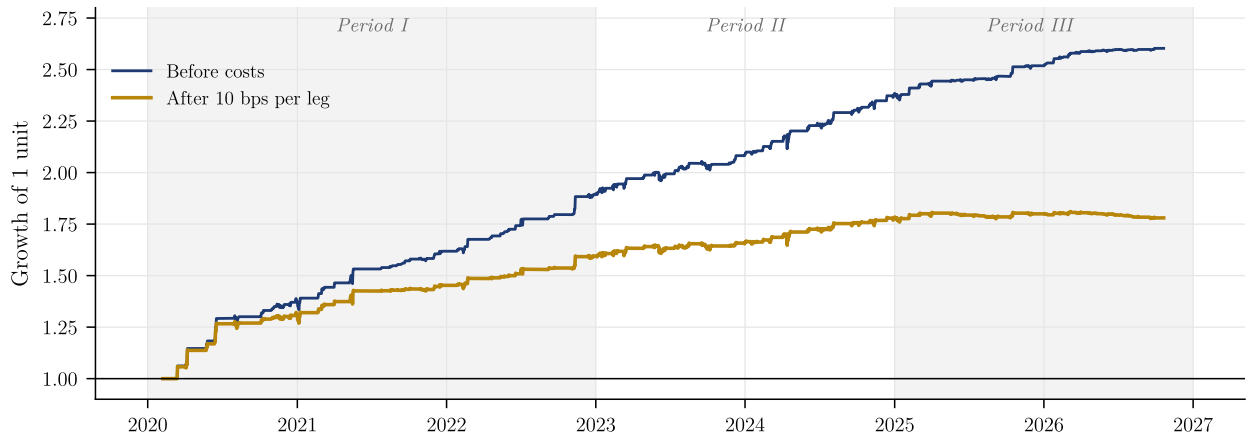


Figure 5. Cumulative growth of one unit, before and after costs. Shading marks the three sub-periods analysed in Section 5. Note the shape of the after-cost line: steep in Period I, gentler in Period II and essentially flat in Period III.

A careful reader will already have spotted the problem in Figure 5. The before-cost line keeps climbing all the way to 2026. The after-cost line does not. Something happened to this trade, and it was not that the mean reversion went away.

5 The Incredible Shrinking Edge — *where the money went*

Table 5 splits the sample into three periods: the token markets' infancy (2020–22), the high-rate years (2023–24) and the most recent 22 months (January 2025 to October 2026). Figure 6 repeats the exercise year by year. The pattern is unambiguous.

Table 5. Alpha decay: spread dynamics and strategy performance by sub-period.

	Period I *2020–2022*	*Period II* *2023–2024*	*Period III* *2025–2026*
Spread dynamics			
Standard deviation of spread	92.1 bps	71.0 bps	28.3 bps
Mean absolute spread	63.7 bps	68.8 bps	29.0 bps
Half-life, $t_{1/2}$	1.54 days	3.58 days	2.63 days
ADF p-value	0.0003	0.0018	0.0005
Trading			
Round trips	44	30	23
Average holding period	4.5 days	8.5 days	11.7 days
Average deviation at entry	119 bps	85 bps	44 bps
Gross profit per round trip	145.5 bps	75.9 bps	38.7 bps
Round-trip cost (4×10 bps)	40 bps	40 bps	40 bps
Performance			
Sharpe ratio, before costs	2.53	1.92	2.02
Sharpe ratio, after costs	**1.88**	**0.93**	**−0.02**
Annualised return, after costs	17.43%	5.73%	−0.06%
Maximum drawdown, after costs	−3.96%	−3.26%	−1.85%
Break-even cost per leg, c^*	**36.8 bps**	**19.6 bps**	**9.9 bps**

Period I begins on 7 February 2020 and Period III ends on 21 October 2026. Round trips are counted from the first day of each period, so a position open at a boundary is counted in both periods.

Three facts stand out. First, *the spread got small*. Its standard deviation fell by more than two-thirds, from 92 to 28 bps, and the typical entry signal now fires at a 44 bps dislocation rather than a 119 bps one. Second, *the spread did not stop reverting*. The ADF test rejects a unit root comfortably in every period, the half-life stays at a few days, and the before-cost Sharpe ratio in Period III (2.02) is barely below that of Period I. A naïve reading of before-cost results would conclude that nothing had changed.

Third, and decisively, *the gaps became smaller than the cost of trading them*. Gross profit per round trip halved and then halved again, from 145 to 76 to 39 bps, against a fixed round-trip cost of 40 bps. In Period III the average trade earns 38.7 bps and pays 40. The break-even cost per leg falls in near lock-step, from 37 to 20 to 10 bps, until it lands almost exactly on the assumed cost. The after-cost Sharpe ratio follows it from 1.88 to 0.93 to −0.02. Figure 6(b) shows the same convergence year by year: the break-even cost crosses below 10 bps for the first time in 2026.

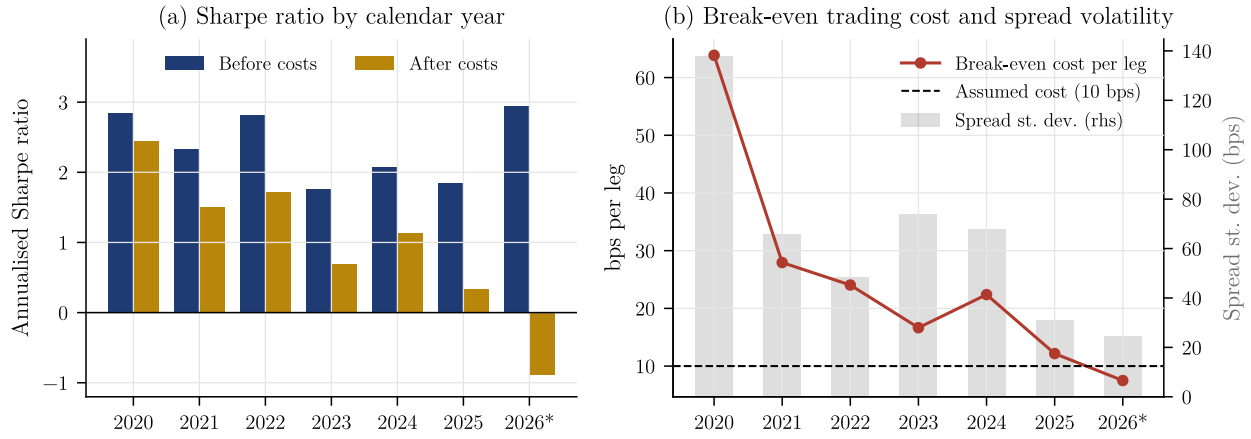


Figure 6. Alpha decay by calendar year. Panel (a): annualised Sharpe ratio before and after costs. Panel (b): the break-even per-leg trading cost, eq. (9), against the assumed 10 bps (dashed), with the standard deviation of the spread on the right-hand axis. An asterisk denotes the partial year (2026, to 21 October).

Two further details from Appendix Table A1 sharpen the story. In 2026 alone the ADF test no longer rejects a unit root at conventional levels ($p = 0.34$) and the estimated half-life lengthens to 5.8 days. With dislocations this small, daily data can no longer reliably distinguish the residual signal from noise. And the holding period has stretched from 4.5 to 11.7 days, so each trade now ties up capital for longer in exchange for a smaller gross reward.

The interpretation is the one the adaptive markets hypothesis would predict (Lo, 2004). In 2020 the token markets were small, fragmented and lightly arbitraged. A two-ounce trade between identical claims could earn more than a percentage point in a few days, because few participants were willing or able to do it. As volumes grew, market makers and cross-venue arbitrageurs moved in and competed the spread down until the marginal opportunity was worth roughly what it cost to take. The limits-of-arbitrage view of Shleifer and Vishny (1997) and the evidence on ETF premiums (Petajisto, 2017) both suggest that this residual band, a few tens of basis points wide, is about as tight as such markets get when transacting has real costs.

In one sentence

The PAXG–XAUT gap still snaps shut within days. It is simply no longer wide enough, after 2024, to pay for the trip.

6 Limitations — *reasons to be less impressed*

The results above are a measurement exercise, not a deployable strategy. Several caveats point in the direction of making the historical profits *smaller*, not larger.

- **Borrow and shorting costs.** Each trade requires short-selling one token, which means borrowing it. Borrow fees, availability constraints and recall risk are not modelled. Any positive borrow cost would make Period III unambiguously loss-making.
- **Closing prices and bid–ask bounce.** Yahoo closes are aggregated across venues and are not executable quotes. Roll (1984) shows that bid–ask bounce alone induces negative autocorrelation in price changes. Some of the measured mean reversion, particularly for the thinner XAUT market in 2020, may be an artefact a trader could not capture.
- **Daily sampling.** Intraday dislocations, which are probably where today's arbitrageurs actually operate, are invisible at a daily frequency. The true half-life is likely shorter and the true opportunity more competitive than daily data suggest.

- **Flat cost assumption.** A constant 10 bps per leg ignores the fact that trading costs themselves fell as markets deepened. If costs fell faster than the spread did, Period III could be less bleak for a low-cost market maker. The break-even cost in Table 5 is reported precisely so that readers can apply their own cost estimate.
- **No optimisation and no out-of-sample test.** The parameters were fixed in advance, which avoids data-mining but also means no attempt was made to find a better rule. A shorter window or a tighter exit might recover some edge. Equally, they might not, and testing many rules on one history would raise its own multiple-comparisons problem.
- **Benchmark choice.** Gold futures are a convenient but imperfect proxy for the spot price the tokens actually reference. A spot benchmark such as the LBMA Gold Price would give cleaner tracking statistics, though it would not change the token-pair results at all.

7 Conclusion

Gold-backed tokens are a natural test of whether tokenisation delivers what it promises. On the evidence of 2020–2026, the answer is *increasingly yes*. Apparent tracking errors against gold are largely an artefact of measurement: asynchronous closes and futures carry. The genuine friction, the gap between two tokens backed by identical ounces, has always been mean-reverting, closes with a half-life of about two and a half days, and has shrunk by more than two-thirds since the market's infancy. A trader exploiting that gap earned a respectable after-cost Sharpe ratio of 1.88 in 2020–22, roughly half that in 2023–24, and nothing at all since 2025.

For researchers, the episode is a compact illustration of alpha decay in a market with unusually clean fundamentals. For institutions and regulators designing frameworks for tokenised real-world assets, it carries a more encouraging message: given time, liquidity and enough participants, a tokenised asset can come to trade very close to the value of what it represents. The arbitrage that made this possible was, for a while, profitable. It is now simply the price of the market working.

The market, it turns out, read the same textbook.

Notes

[1] PAXG is issued by Paxos and XAUT by Tether. Both are redeemable, subject to each issuer's terms and minimum sizes, for allocated physical gold. Redemption minimums and fees are one plausible reason why small deviations are not arbitraged away instantly.

[2] Ornstein and Uhlenbeck developed the process in 1930 to describe the velocity of a particle buffeted by molecular collisions. That it describes two gold tokens on a blockchain almost a century later says either something deep about the universality of mean reversion or something about the limited imagination of quantitative finance. Possibly both.

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Appendix A Year-by-year results

Table A1. Spread dynamics and strategy performance by calendar year.

Year	*Spread s.d. (bps)*	*Half-life (days)*	*ADF p-value*	*Round trips*	*Net return*	*Sharpe (gross)*	*Sharpe (net)*	*Break-even cost (bps/leg)*
2020[a]	137.9	1.40	0.058	13	30.9%	2.84	2.44	63.9
2021	65.8	1.80	0.003	15	11.0%	2.33	1.50	27.9
2022	48.5	1.58	0.001	17	9.7%	2.81	1.72	24.0
2023	73.9	4.77	0.006	15	3.8%	1.76	0.70	16.7
2024	67.7	2.70	<0.001	16	7.7%	2.08	1.13	22.4
2025	31.1	2.00	0.005	12	1.0%	1.85	0.33	12.2
2026[b]	**24.4**	**5.82**	**0.343**	**11**	**−1.1%**	**2.94**	**−0.88**	**7.5**

[a] From 7 February 2020. [b] To 21 October 2026. Net return is the cumulative after-cost return within the year. Positions open at a year boundary are counted in both years. The 2020 ADF statistic narrowly misses the 5% level despite the fastest half-life in the sample, reflecting a few extreme March 2020 observations and the short sample.

Appendix B Reproducibility

The analysis is a single Python script using `yfinance`, `pandas`, `numpy`, `statsmodels` and `matplotlib`. It downloads the data, computes every statistic in this paper, runs the backtest and produces the figures. All parameters are set once at the top of the script: the 30-day window, ±2 entry and 0 exit thresholds, and 10 bps per leg. The backtest applies each day's signal from the following day only, so no position uses information unavailable at the time. Because Yahoo Finance revises and extends its history, re-running the script at a later date will change the final digits of some results, but not the conclusions.

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